



POLITECNICO

MILANO 1863

RF Systems Project 2025-2026

Prof. OLDONI MATTEO

Team member

Hariharan Murugan Marichamy - 11047334

Balasakthi Shanmugaraja – 11033318

Karuniaa Jacob Selvi – 11034239

RF front-end of a conversion receiver

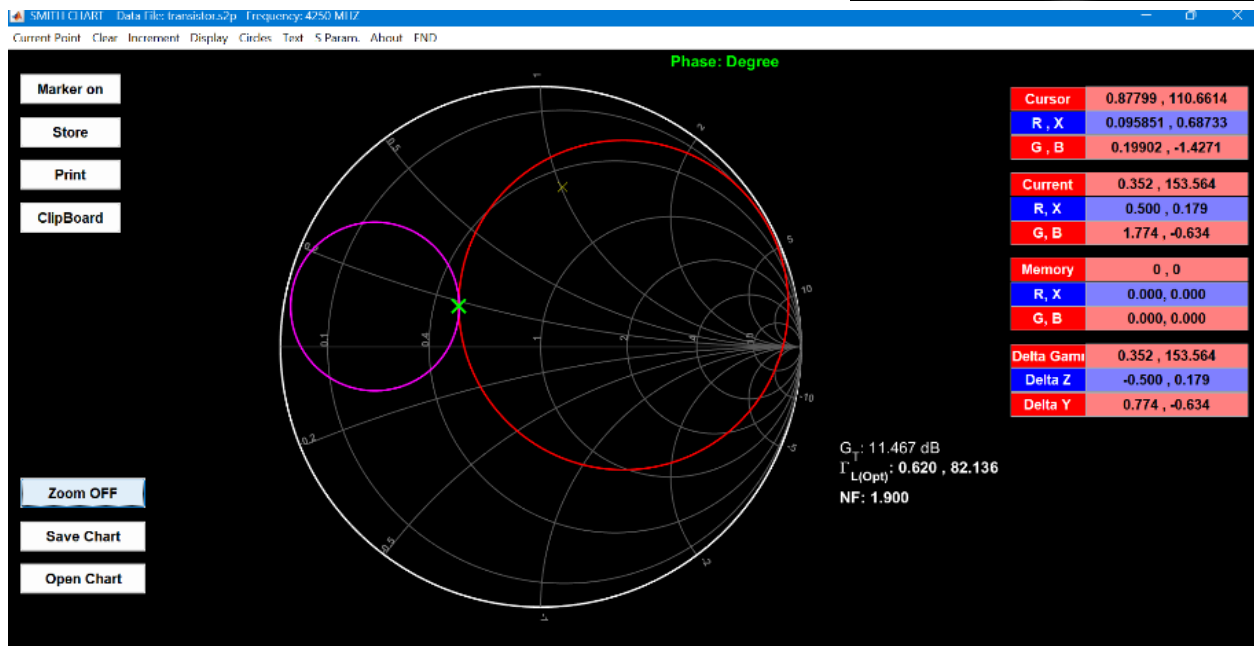
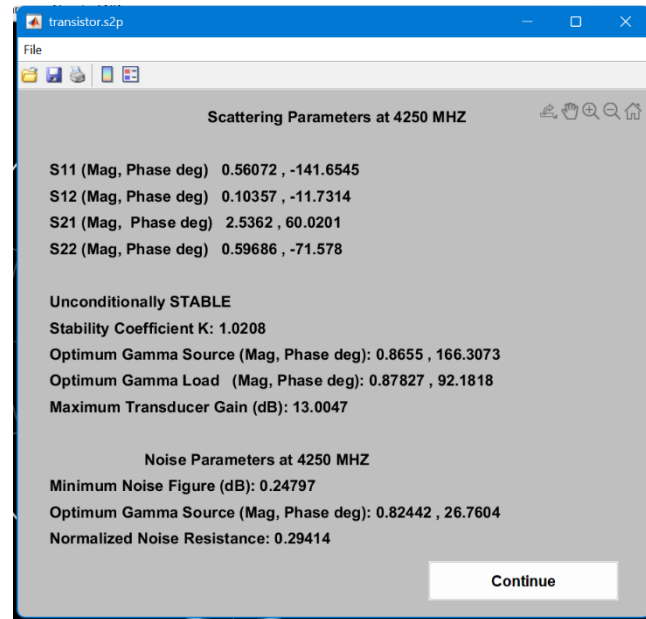
This project evaluates the performance of the **RF front-end of a conversion receiver** through detailed circuit-level and system-level simulations. The RF front-end plays a key role in determining the quality of the received signal and directly affects the overall receiver performance. Using the provided **VSS system diagram**, the main parameters of the RF blocks namely the **Low Noise Amplifier (LNA)**, **mixer**, and **Local Oscillator (LO)** are identified and analyzed. While some parameters are predefined, others are **extracted from circuit simulations**, reflecting practical design and analysis conditions. The extracted parameters include **gain**, **noise figure**, **IP3**, and **1-dB compression point**, which are then used to evaluate system-level performance. The analysis further focuses on **Error Vector Magnitude (EVM)**, considering the effects of **phase noise** and **image interference**, in order to assess the limitations and dynamic behavior of the RF front-end.

LNA:

Transistor Characterization Using S-Parameters and Smith Chart

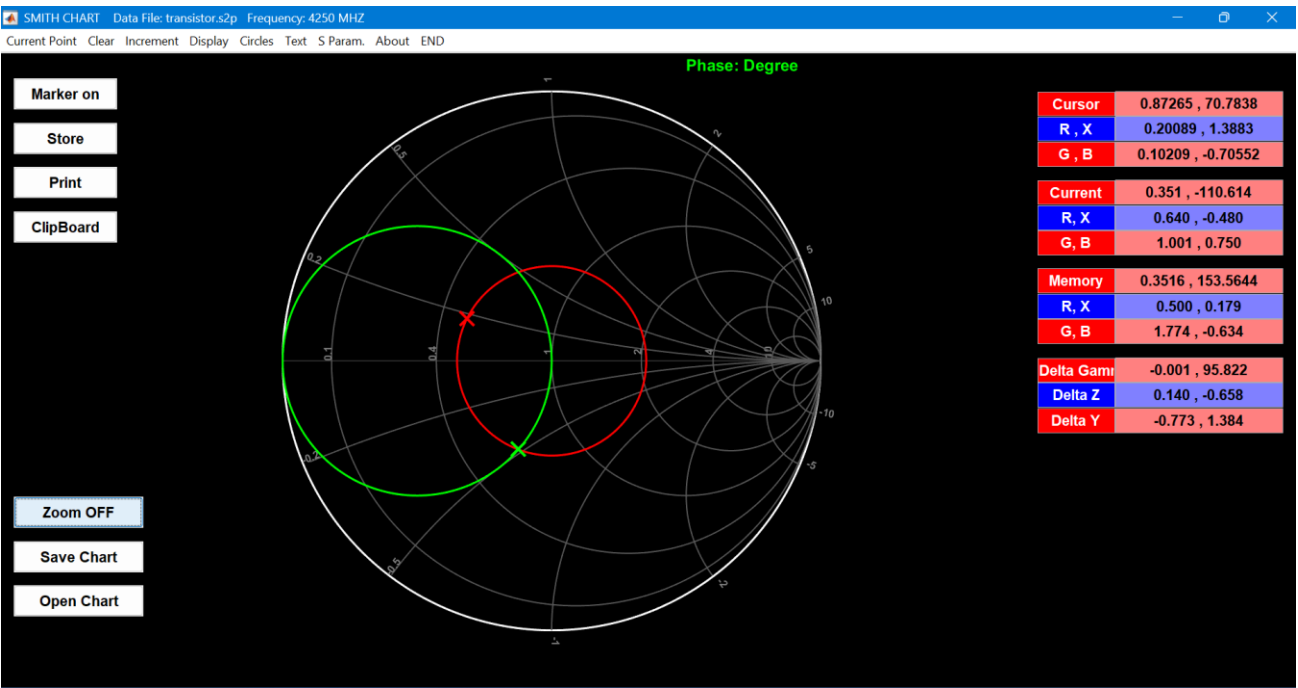
The transistor provided in **S-parameter format (transistor.s2p)** was analyzed using the **Smith Chart tool** at the operating frequency of **4.25 GHz**. After loading the S-parameter file and setting the frequency, the Smith Chart window displays the transistor input and output behavior together with key performance indicators such as gain, optimum source reflection coefficient, and noise figure. The device was verified to be **unconditionally stable** at the design frequency.

The first circle drawn on the Smith Chart is the **noise circle**, corresponding to a **noise figure of 1.95 dB**. After identifying the noise circle, several **available gain circles** were plotted. Among them, an **available gain of 11.46 dB**



was found to align well with the noise circle from **S-parameter** , the **optimal Gamma load($\Gamma_{L,opt}$)** was obtained, as shown in the figure

Figure X



From the **Smith Chart cursor readings** shown in **Figure X**, the $\Delta\Gamma$ **phase shift** was observed to be **95.8°**, which was rounded to **96°**. Since this phase variation corresponds to **twice the electrical rotation** on the Smith Chart, the required **electrical length** was taken as **48°**. This value was entered in the **TXLINE microstrip model** in the **AWR design environment**, as shown in **Figure Y**. The **dielectric constant**, **substrate height**, **conductor thickness**, and **line width** were kept exactly as specified in the project file.

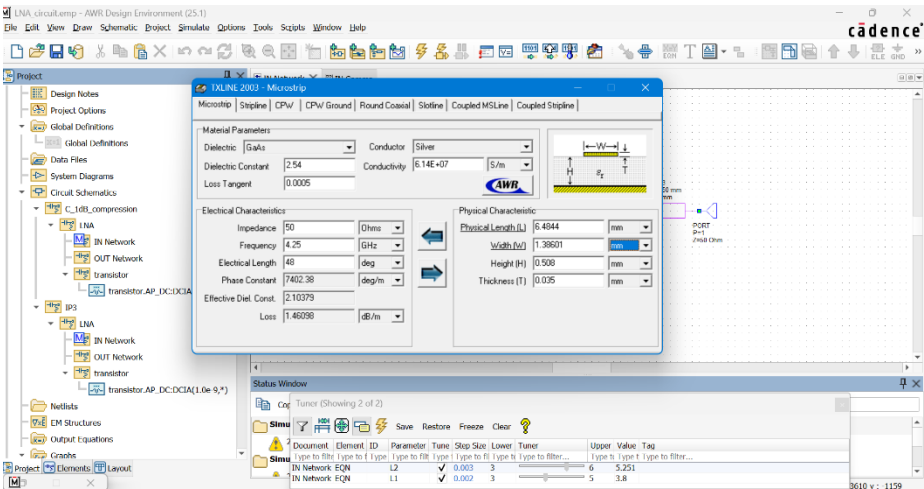


Figure Y

After converting this electrical length into a **physical length** using TXLINE (Figure Y), a second electrical length was obtained from the Smith Chart by taking the $\tan^{-1}$ of the **susceptance value**, where **B = 0.750**, resulting in an electrical length of approximately **36.8°**. This value was also implemented in the TXLINE microstrip model to obtain the corresponding physical length, as shown in **Figure Z**.

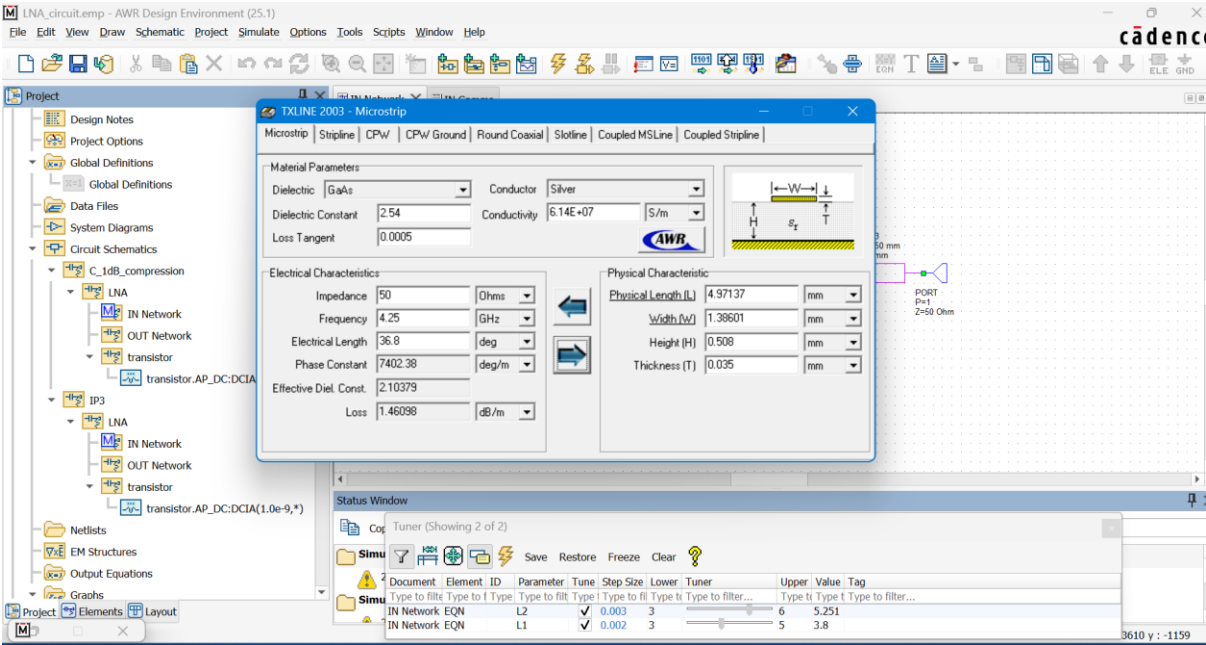


Figure Z.

As a result, two transmission line lengths, **L1** and **L2**, were obtained. Using these two lengths, the **input reflection coefficient (Γ_{in})** was evaluated in the LNA schematic. The values of L1 and L2 were then **tuned** until the simulated Γ_{in} matched the Smith Chart result, with a magnitude of approximately **0.35** and a phase of about **153°**As shown in **Figure W**.



Figure W.

Once the matching condition was satisfied, the **LNA circuit schematic** was simulated, and the required performance graphs were obtained for further analysis.

Simulations and results:

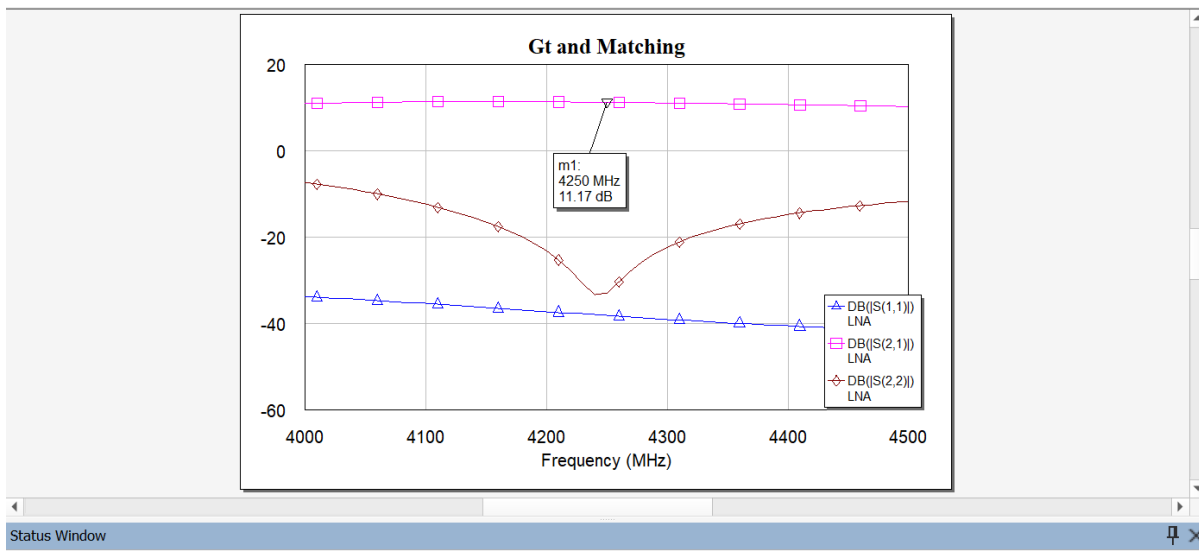


Figure :Gt and matching

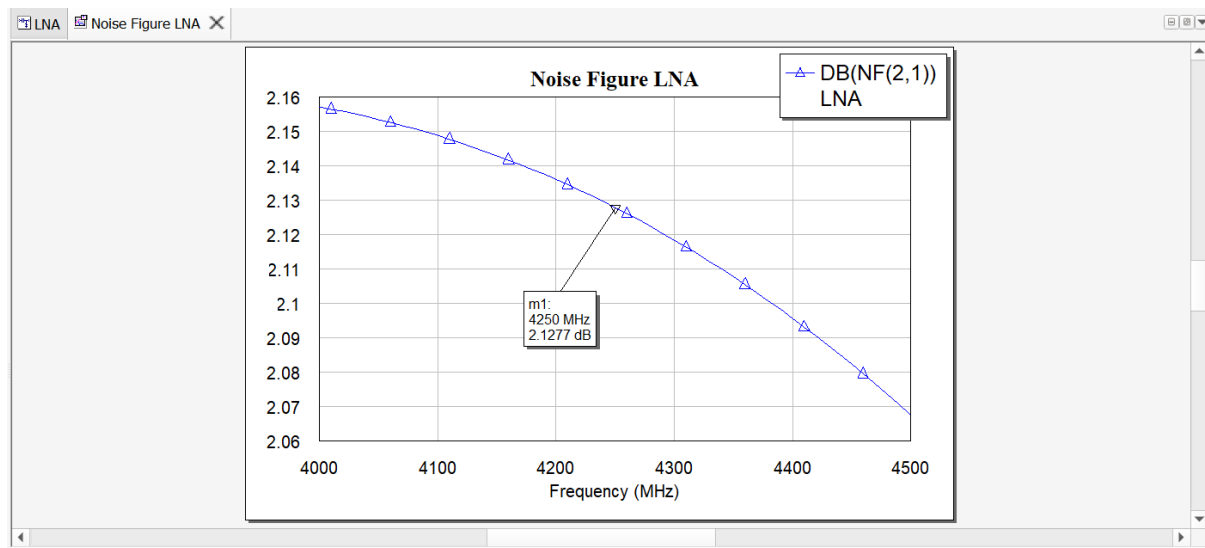


Figure :Noise Figure LNA

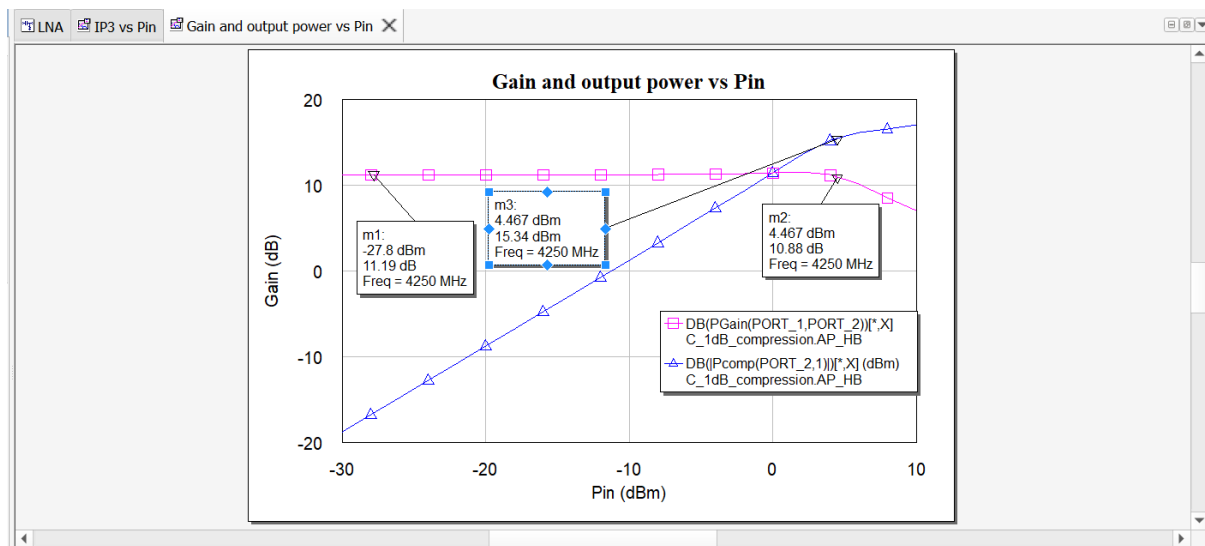


Figure :Gain & output Power Vs Pin

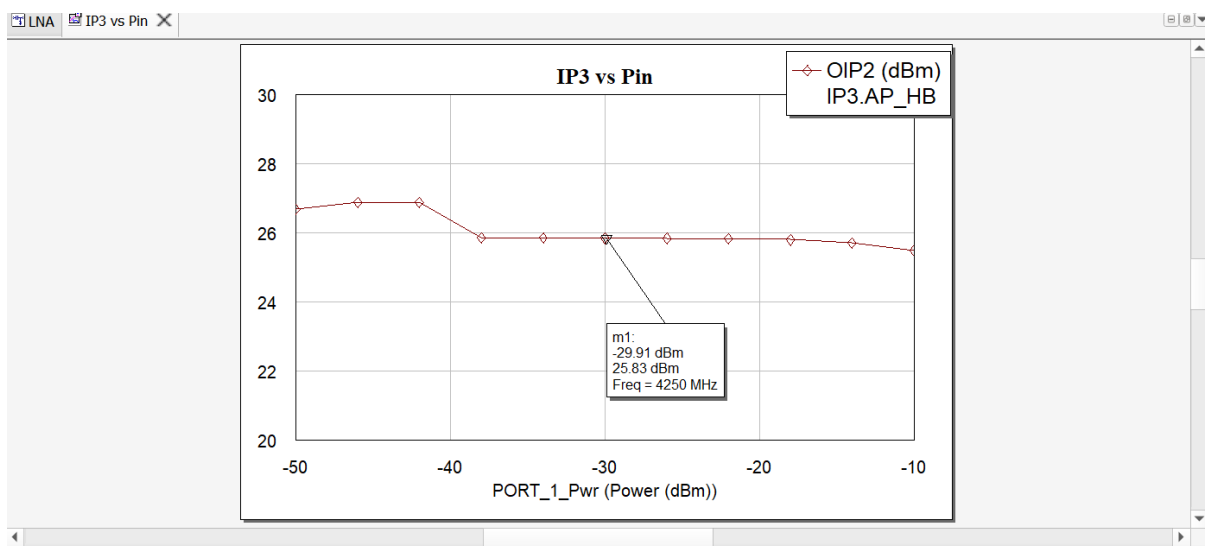


Figure :IP3 Vs Pi

LNA parameters from circuit simulations

- **Gain LNA (Gt and matching)= 12.8 dB**
- **Noise Figure LNA = 2.98 dB**
- **P1dB_LNA (Gain & output Power Vs Pin)= -18.9 dBm**
- **IP3_LNA (IP3 Vs Pin)= 27.4 dBm**

MIXER:

Circuit evaluation of Mixer parameters

The mixer circuit is implemented using a **balanced diode configuration** with **two diodes**, a **rat-race hybrid**, and an **IF filter**, as defined in the file *Mixer_circuit.emp*. All circuit elements are assigned as given and are not modified. The IF filter is included in the schematic for simulation purposes, although it is not considered part of the mixer itself.

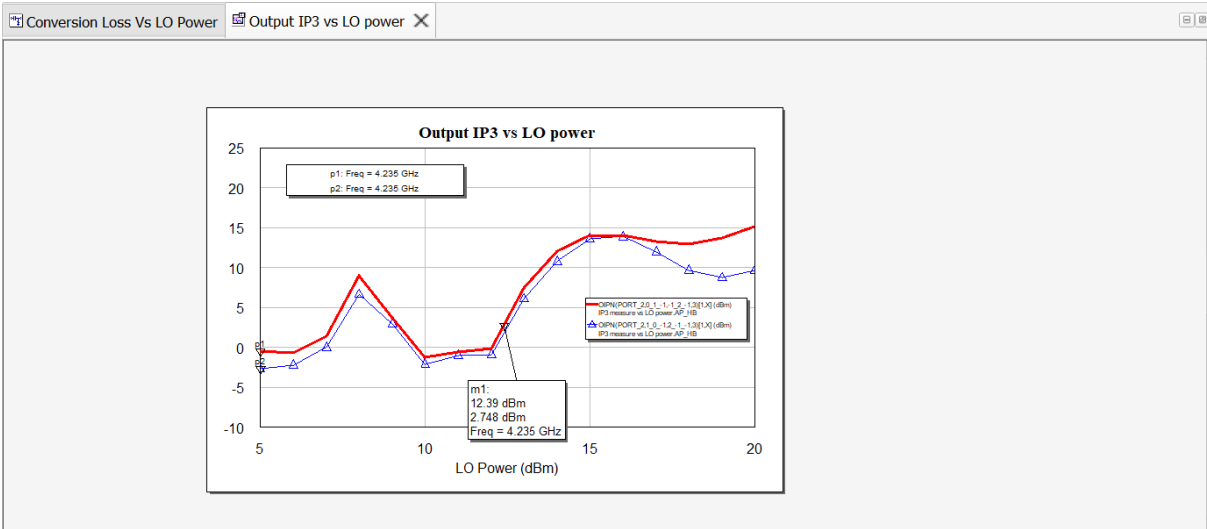
To extract the mixer parameters, simulations were first performed by simulating the circuit schematics of **conversion loss vs LO power** and opening the **conversion gain vs LO power** the **output IP3 vs LO power** graphs were obtained. From these two graphs, the **LO power** was identified, which corresponds to an LO power of approximately **15 dBm**. This value represents the nominal LO drive level that provides a good trade-off between conversion gain and linearity.

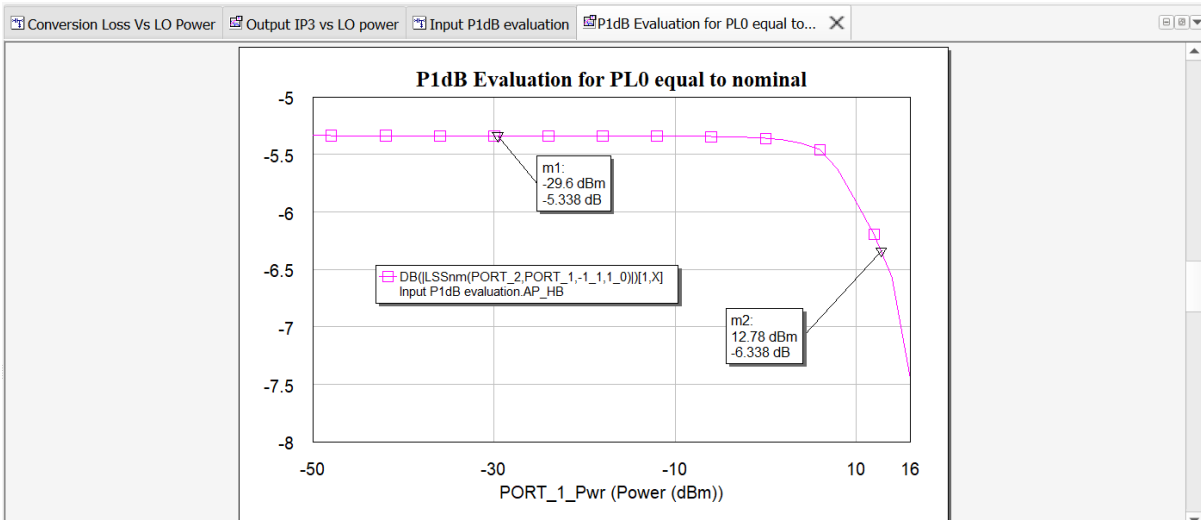
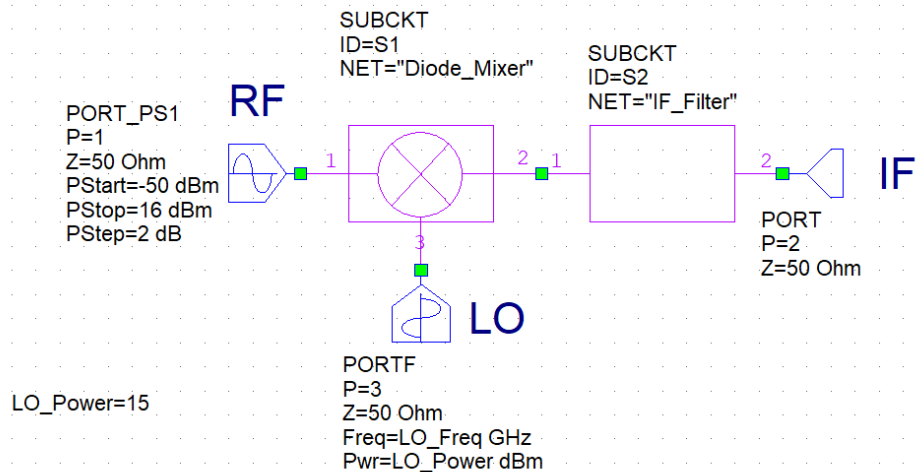
Once the nominal LO power was identified, this value (**LO power = 15 dBm**) was assigned in the **Input P1dB evaluation** schematic. The simulation was then repeated to obtain the **P1dB Evaluation for PL0 equal to nominal** graph of the mixer at the selected LO power. In addition, the **noise temperature vs LO power** graph was obtained.

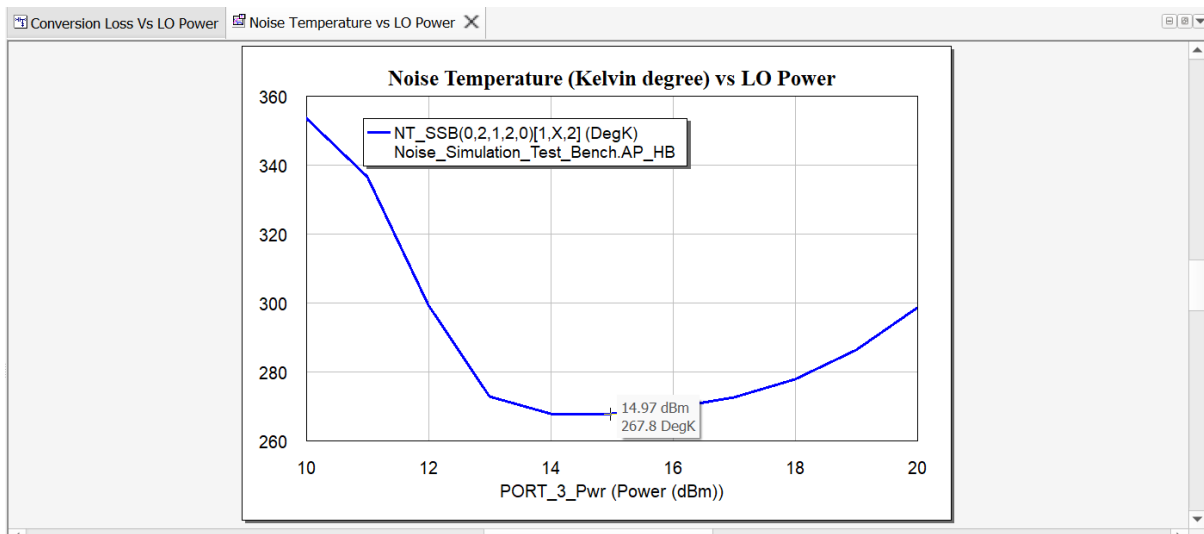
Using these simulations, the required mixer parameters **conversion gain**, **input IP3**, **P1dB**, and **equivalent noise temperature (TSSB)** were extracted from the corresponding graphs at the chosen LO power. These values were then used for the system-level simulations of the RF front-end.

Simulation and Results:

Below are the simulation results obtained for the mixer under the selected operating conditions.







Mixer parameters from circuit simulations

- **Conversion_gain_mixer** (Conversion Gain Vs Lo Power)= -5.3 dB
- **TSSB_mixer**(Noise temp (kelvin degree Vs Lo Power)) = 268 K
- **Input_IP3_mixer** = 14- conversion gain mixer=**19.3 dBm**
- **Input_P1dB_mixer** (P1db Evaluation for PLo equal to nominal)= 12.78 dBm

Error Vector Magnitude (EVM)

Recall of Error Vector Magnitude (EVM):

In digital communications the received signal can be represented by a vector that may assume a limited number of positions. The set of possible positions defines the ideal constellation diagram. Any disturbance overlapping the signal produces a displacement between the ideal constellation and the one generated by the actual received signal. Such displacement represents an indicator of the signal quality degradation due to various sources (including the non-linearity). The parameter used for quantifying the above displacement is the error vector magnitude (EVM).

Definition of EVM:

Given the error vectors between the ideal constellation and the degraded signal, EVM is defined as follows:

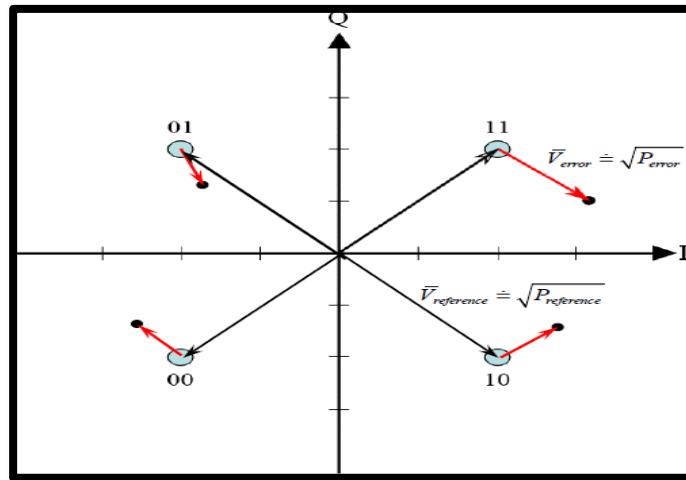


Figure: Constellation of 4-QAM modulation

$$EVM(\%) = 100 \cdot \sqrt{\frac{P_{error}}{P_{reference}}}$$

Error = average power of the error vectors

Preference = average power of the reference vector

$$EVM(dB) = 20 \log_{10} \sqrt{\frac{P_{error}}{P_{reference}}}$$

Quality Requirement and assessment of the received signal power:

- The quality of the signal converted at IF is affected by noise and distortion produced by the front-end.
- An index of the quality degradation is the Error Vector Magnitude (EVM), measured on the IF signal. It is here assumed that the maximum admissible EVM due to the front end alone is 5% (with external contributions, this limit increases to 5.2%).
- The power level of RF signal at the input of the front-end determines the EVM.
- Using VSS, it is requested to find the minimum and maximum power of the signal at the input of the RF front-end within of which EVM is not larger than 5% (or 5.2% with external sources of quality degradation)

Measure of EVM:

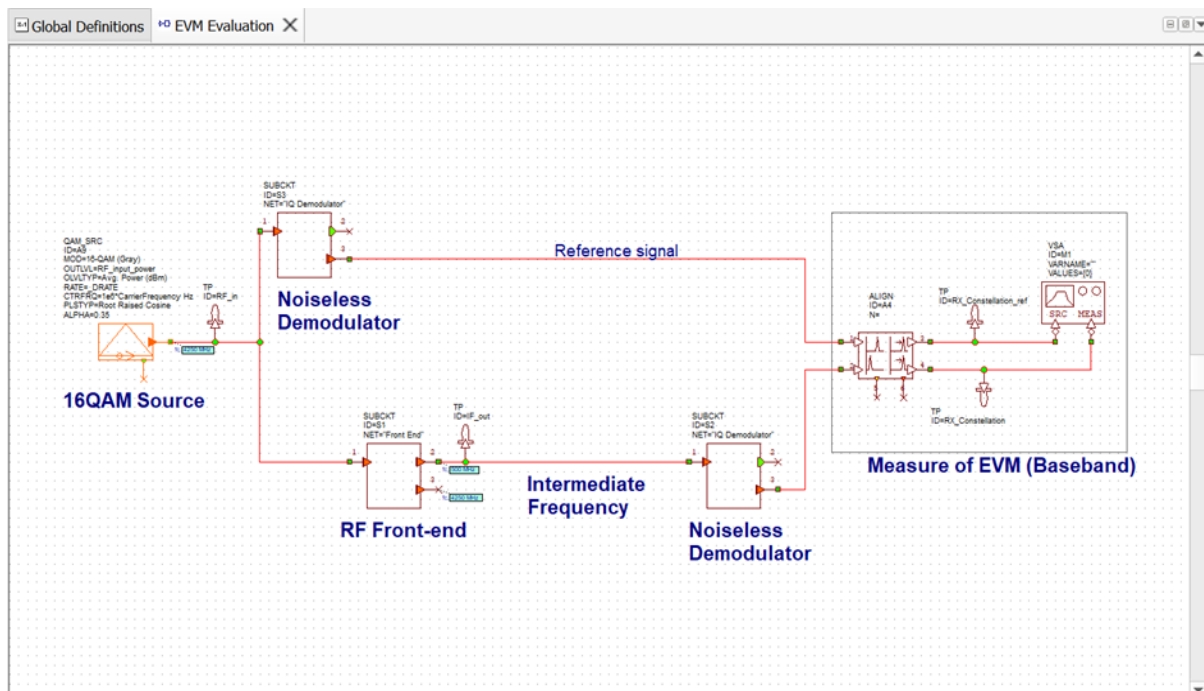


Figure: EVM Evaluation - 16QAM Source

- Both EVM and Constellation measurements are available with VSS simulation. The global variable RF Input Power is varied starting from a very low value (e.g. -100 dB). When EVM arrives to 5% the value of the variable represents the minimum admissible power (due to the noise produced by LNA, Mixer and Filters).
- Further increasing the input power causes EVM to decrease until the non-linearity effects (due to LNA and Mixer) degrades the signal and EVM starts to increase again. We keep increasing the input power until EVM is again 5%. The value of RF_input_power is the maximum admissible power.

Simulation & Result

In the RF front end file we open the Global Definitions window and changed the parameter of LNA and Mixer and the LO_power. And select the variable RF_input_power for tuning. By opening the graphs “Constellation” and “EVM”, we can track the variation of EVM by varying the variable RF_input_power. Then by opening the tuner and changing slowly the value of RF_input_power with the slider (or writing the value in the box). After each change, we wait until the new values appear. then we take as result of the measure the mean value (about) of the fluctuations.

We obtained the Minimum Admissible power and Maximum Admissible power equal to **-74.5** and **4.15** respectively.

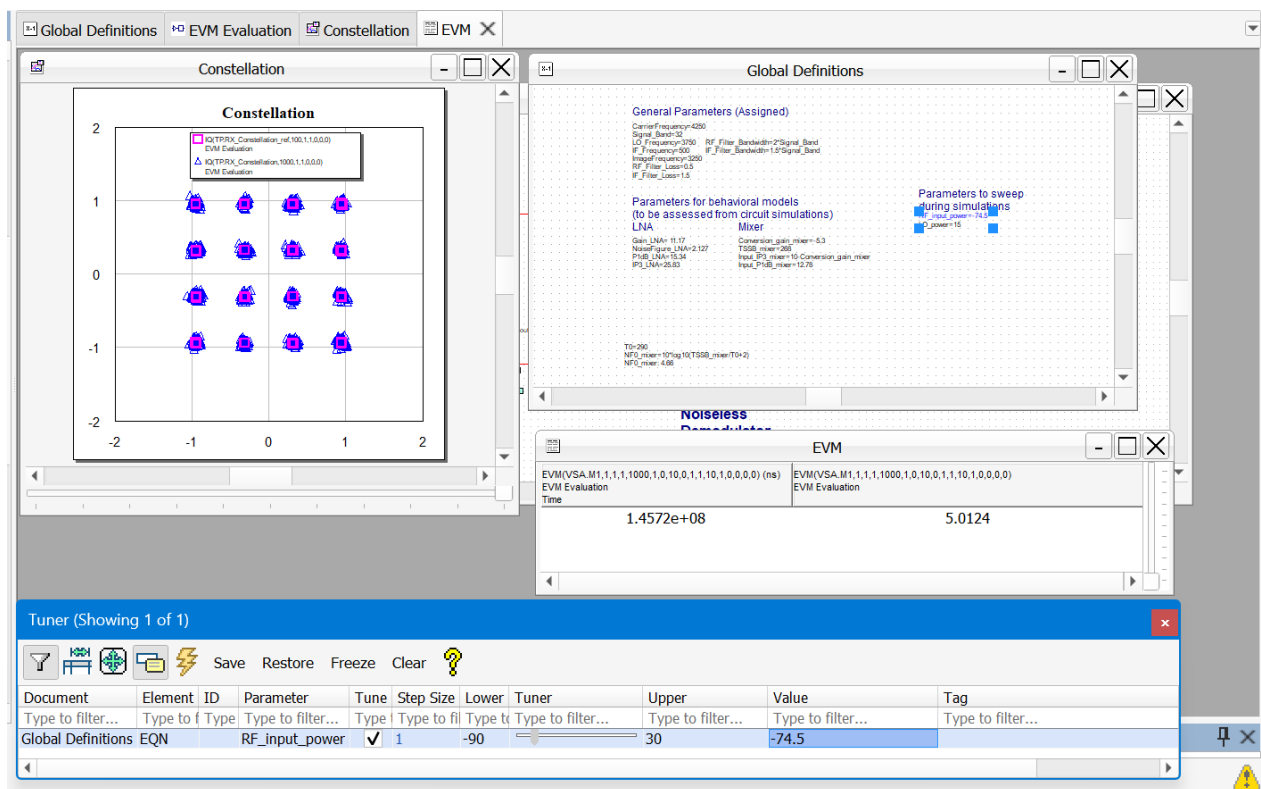


Figure: Constellation and EVM Value of Minimum Admissible Power

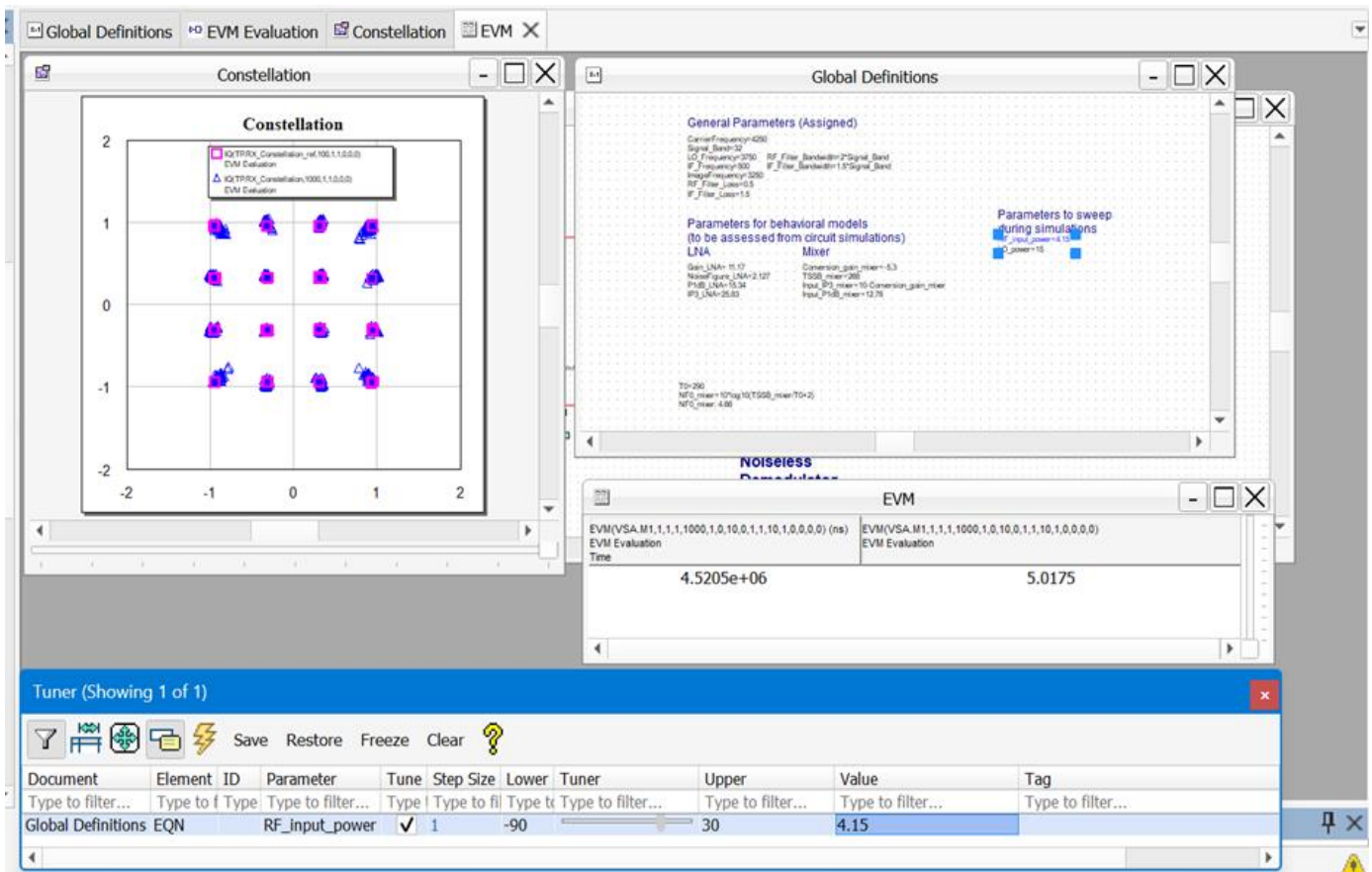


Figure: Constellation and EVM Value of Maximum Admissible Power

The Reason why we have this min and max admissible power

As in the Project description explained Rf Input Power has effect on the value of EVM. And by increasing Rf Input Power we see the reduction in EVM amount, because the displacement between the ideal constellation and the one produced by the actual received signal becomes less, so here our objective to find the Minimum and Maximum Admissible Power by changing the Rf Input Power and tracking the EVM Value.

At the end we have Data Range as the following:

$$\text{DataRange} = P_{\text{max}} - P_{\text{min}}$$

$$\text{DataRange} = 4.15 - (-74.5)$$

$$\text{DataRange} = 78.65$$

Recall of Phase Noise

- Phase noise (PN) represents the effects of the instantaneous phase fluctuations (jitter) of the sinusoidal signal produced by a real oscillator.
- When the oscillator affected by PN is used in a mixer, the random variations of the LO phase are transferred to the carrier frequency of the translated signal. When this signal is demodulated, the extracted information is distorted (quality degraded)
- The degradations produced depend on the maximum deviation of the instantaneous phase of the carrier. They are independent on the power of the RF signal
- Phase noise is practically defined by means of a mask representing the relative level of the power density produced by the oscillator at a distance Δf from the carrier frequency, for various values of Δf .

Choice of the Local Oscillator phase noise mask

- In the previous simulations, the Local Oscillator (LO) is assumed ideal (no phase noise included).
- The acceptable level of Phase Noise is the one not modifying the performances obtained by the RF front- end with no PN present.
- To find this level (defined by the associated Mask), we consider the input RF power equal to the minimum value before found and evaluated EVM.
- If the measured EVM, including PN, is above 5% the phase noise is not negligible, and another mask is required. Continue to reduce PN level until a mask is found with which the measured EVM is about 5% (i.e., it fluctuates between 4.9% and 5.1%). This is the required phase noise mask for the LO.

Simulations & Results

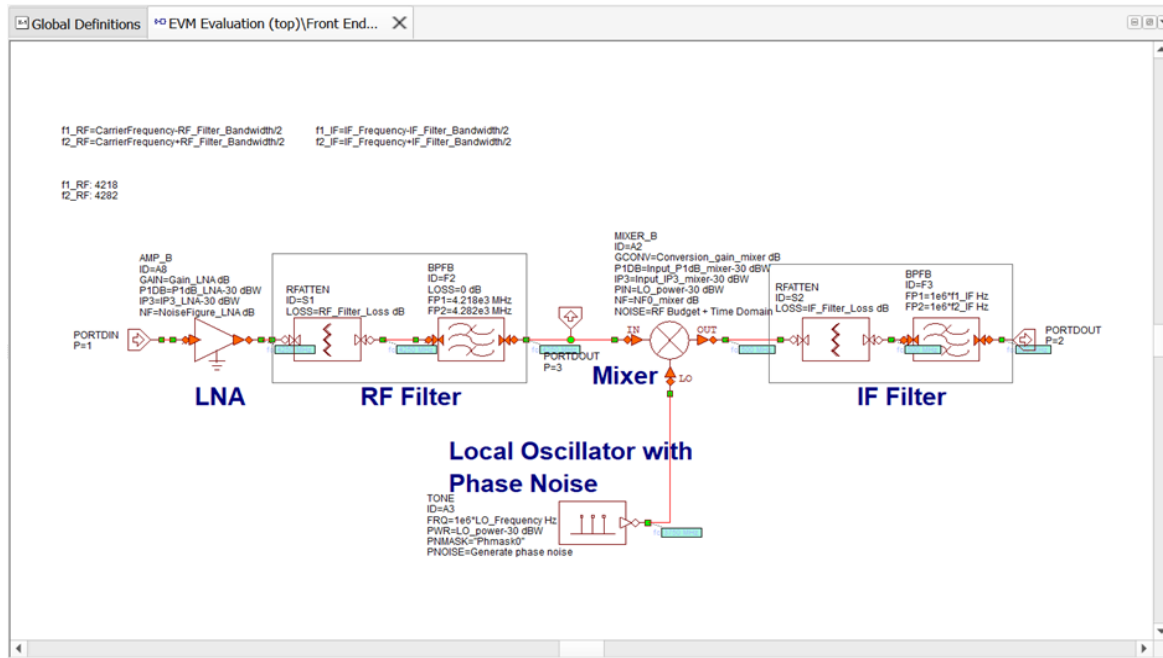
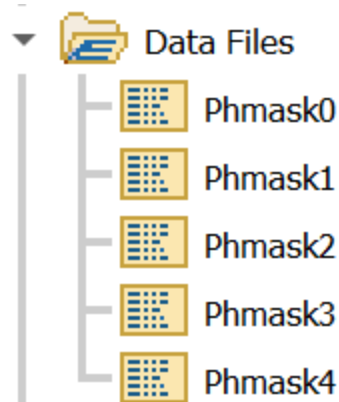


Figure: Evm Evaluation (Front End)

- we changed the “Front End” diagram by select the parameter “PNOISE” of the LO source and set to “Generate phase noise”
- Also, we can change the desired Phmask from 0 to 4 (we have 5 different Phmasks)
- Proceeding with EVM evaluation as previously done and explained.
- 5 phase noise masks are included in the diagram. The masks are defined in the “Data Files” folder and are named Phmask0 to Phmask4 as shown in below.

Data Files of different Phmasks



They are sorted in decreasing phase noise level.

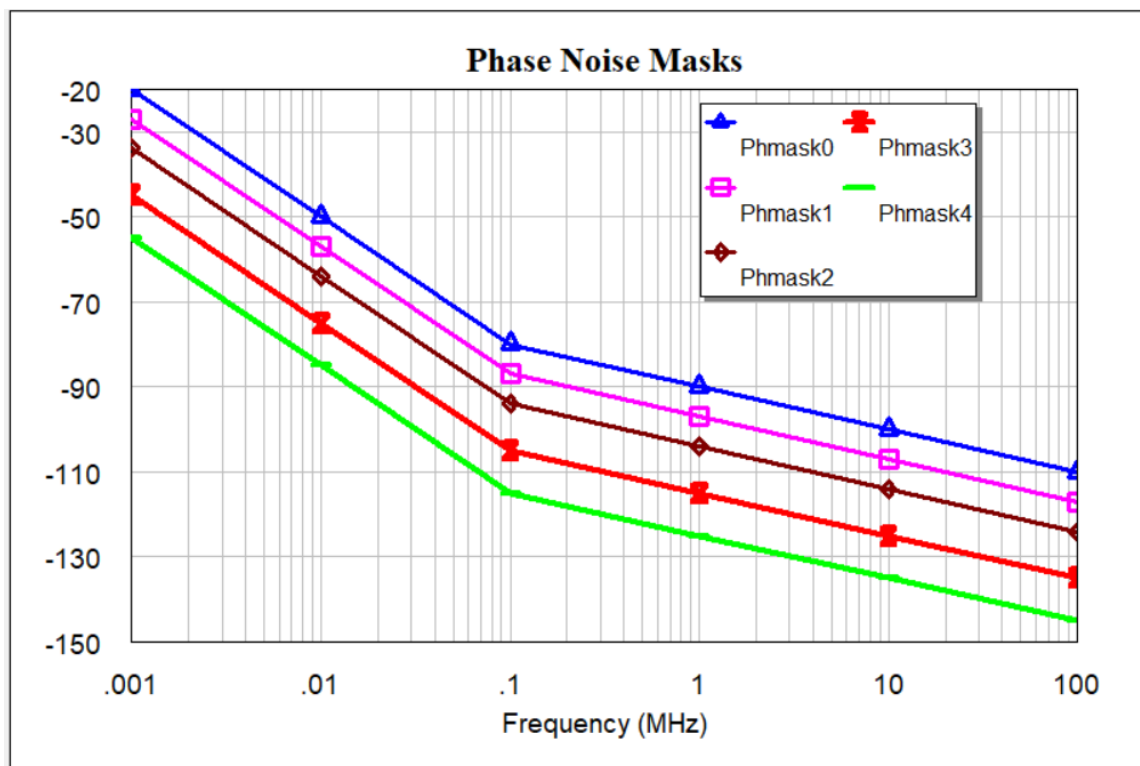


Figure :Phase Noise Level (Decreasing Sort)

To select a mask, use the parameter “PNMASK” of the LO source.

We selected Phmask3. The EVM value of Phmask0, Phmask1, Phmask2, Phmask3, Phmask4 are 17.23, 8.093, 5.8621, 5.083, 5.053 respectively. Both Phmask3 and Phmask4 are suitable for us in terms of EVM value, but we choose Phmask3, because Phmask is expensive.

Below are the result of each phase mask as the following:

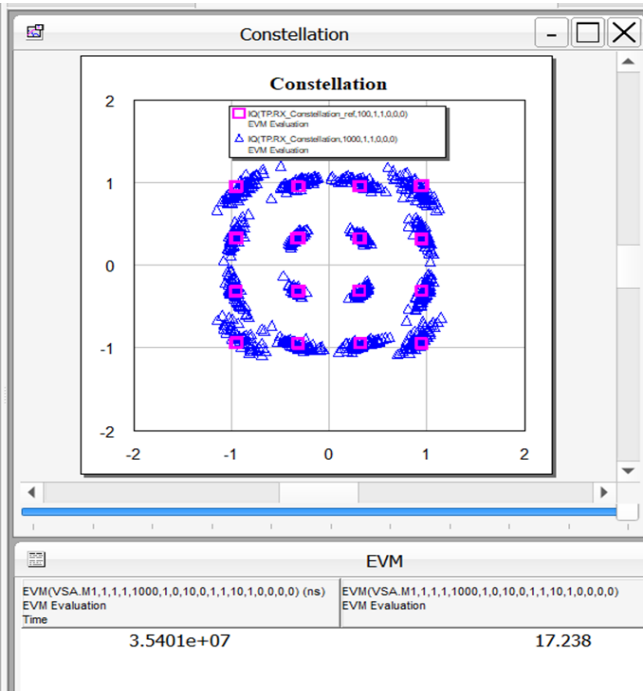


Figure:Constellation and EVM Value by using Phmask0

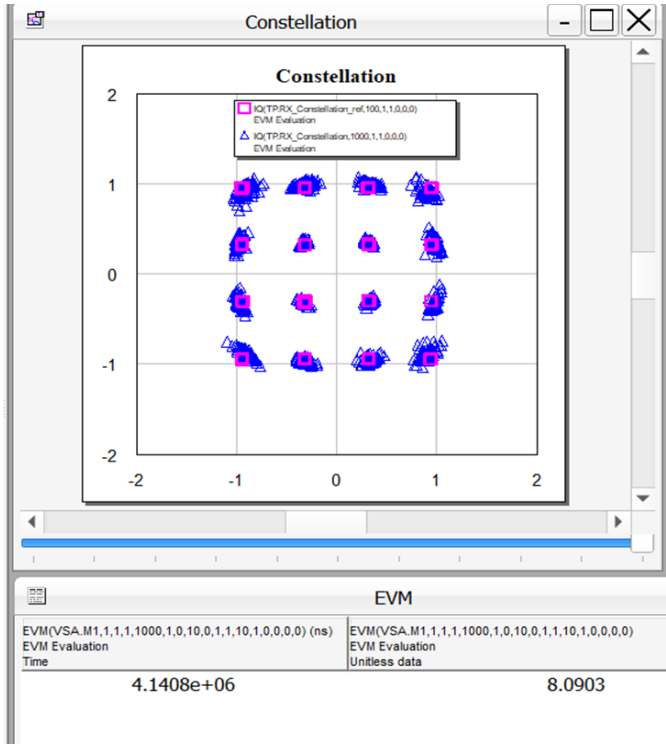


Figure: Constellation and EVM Value by using Phmask1

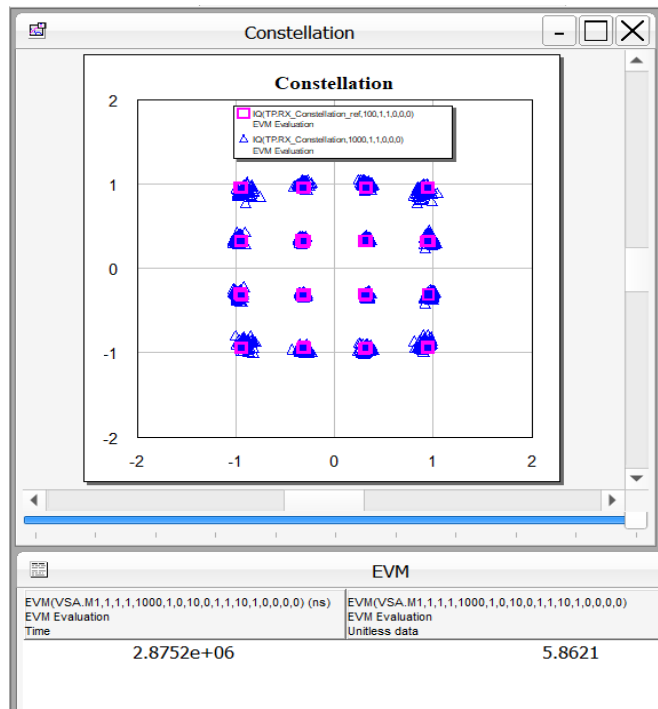


Figure: Constellation and EVM Value by using Phmask2

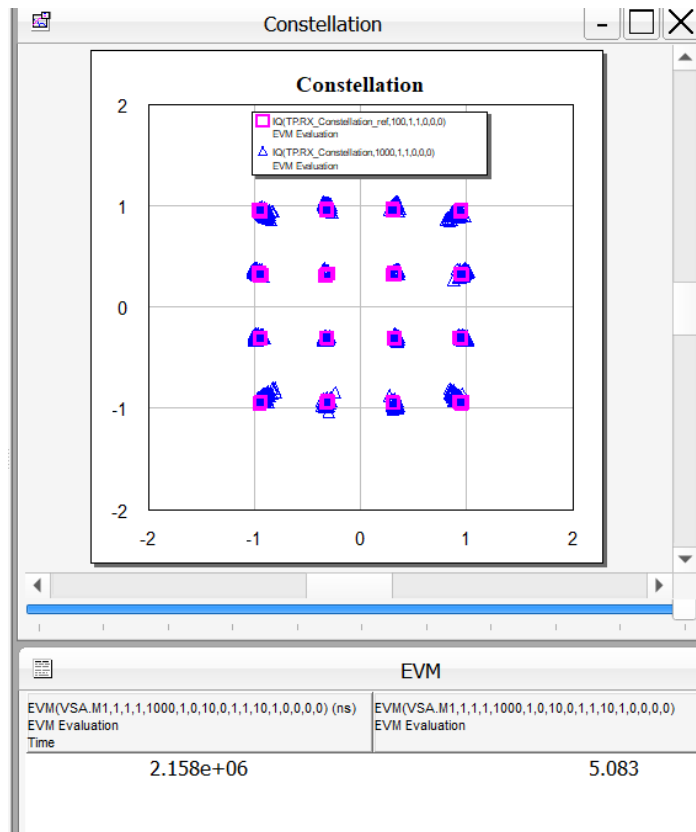


Figure: Constellation and EVM Value by using Phmask3

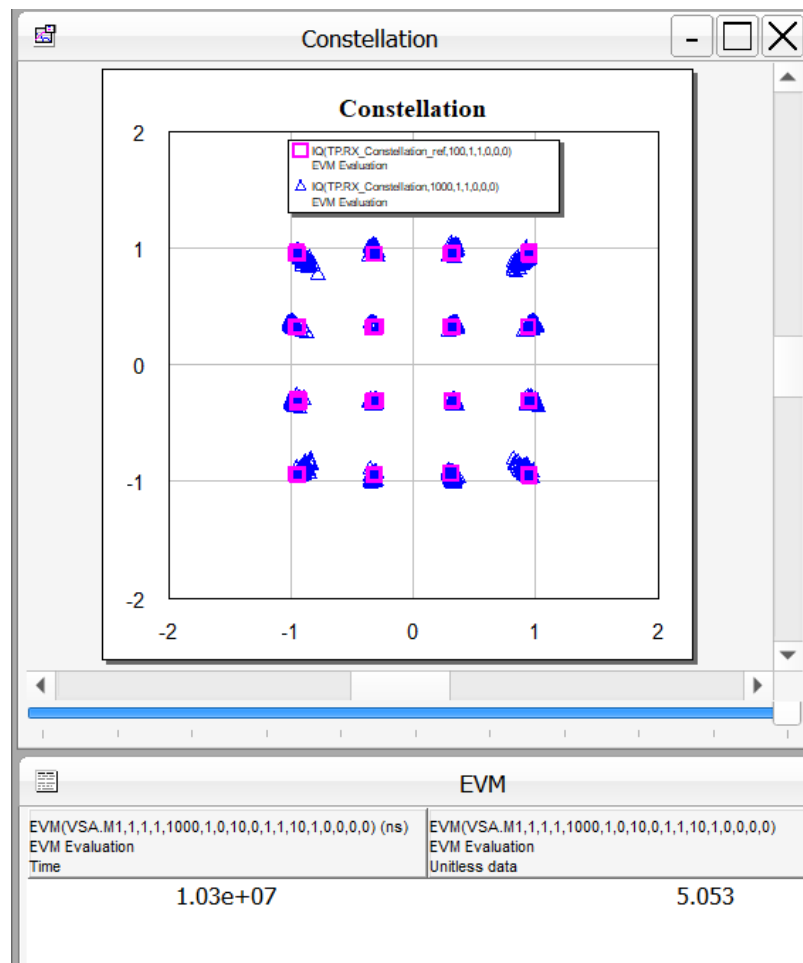


Figure: Constellation and EVM Value by using Phmask4

Degradation due to an image interferer

- An image interferer is a signal at the input of the RF frontend, whose carrier frequency is equal to the image frequency of the mixer.
- This interferer should be eliminated by the image filter. In the practice, it appears at the input of the mixer, with a level depending on the filter selectivity.
- Even if the attenuation of the image filter is large, when the received RF signal is much weaker than the interferer level, the converted signal at IF is distorted. The quality of the demodulated signal at baseband is then degraded.

Evaluation of the interferer maximum power

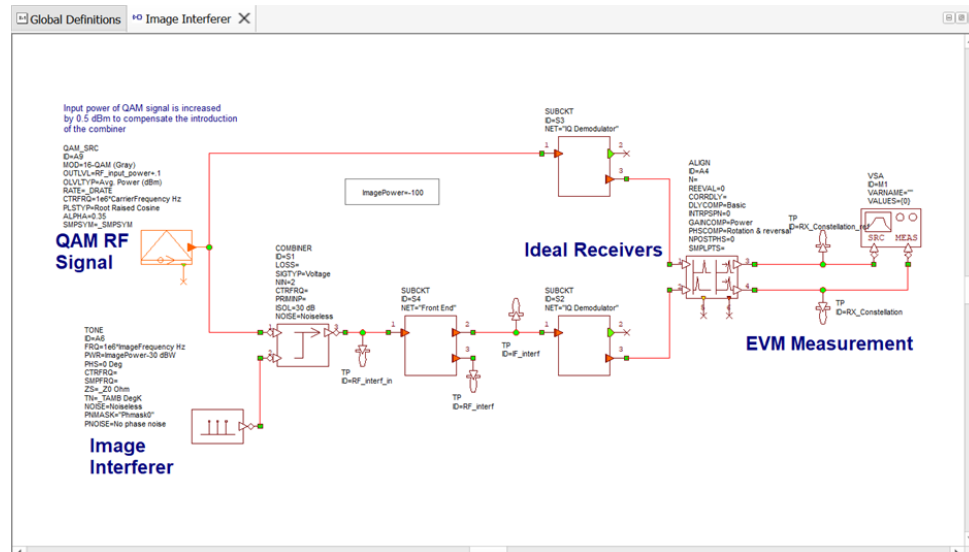


Figure : Image Interferer Scheme

- The interferer is represented by a single tone with the power defined by the variable ImagePower (defined in the diagram)
- The power is initially set to -100 dBm and RF_signal_power to the minimum value previously found. The resulting EVM should be around 5%
- The goal is to find the maximum value of ImagePower for which EVM is equal to about 5.2% (including Phase Noise due to LO)
- In the diagram “FrontEnd” that Phase Noise generation was enabled in the LO source (with the mask before selected which was Phmask3)
- Before starting the simulation, I enable the measurements of the graphs “EVM_Image Interferer” and “Constellation_Image Interferer” also RF Spectrum 1 and 2
- At first, I assigned ImagePower=-100 and started the simulation, while checking the EVM value does not exceed 5%.
- Then I continued increasing ImagePower until EVM arrives to 5.2%. The value of ImagePower represents the maximum admissible power of the interferer.
- We reached 5.2% of EVM by ImagePower = -74dB

Simulations & Results:

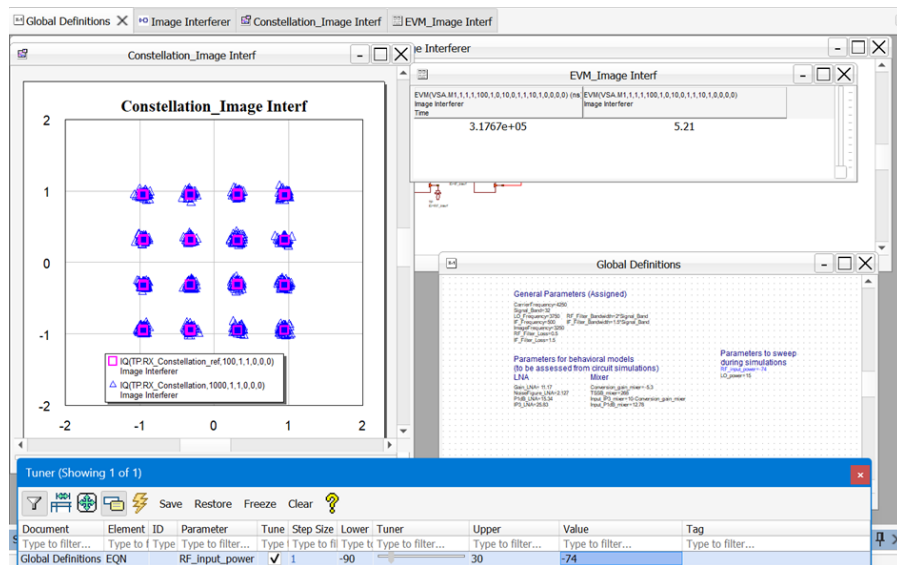


Figure: The Constellation & EVM Value of the Image Interferer

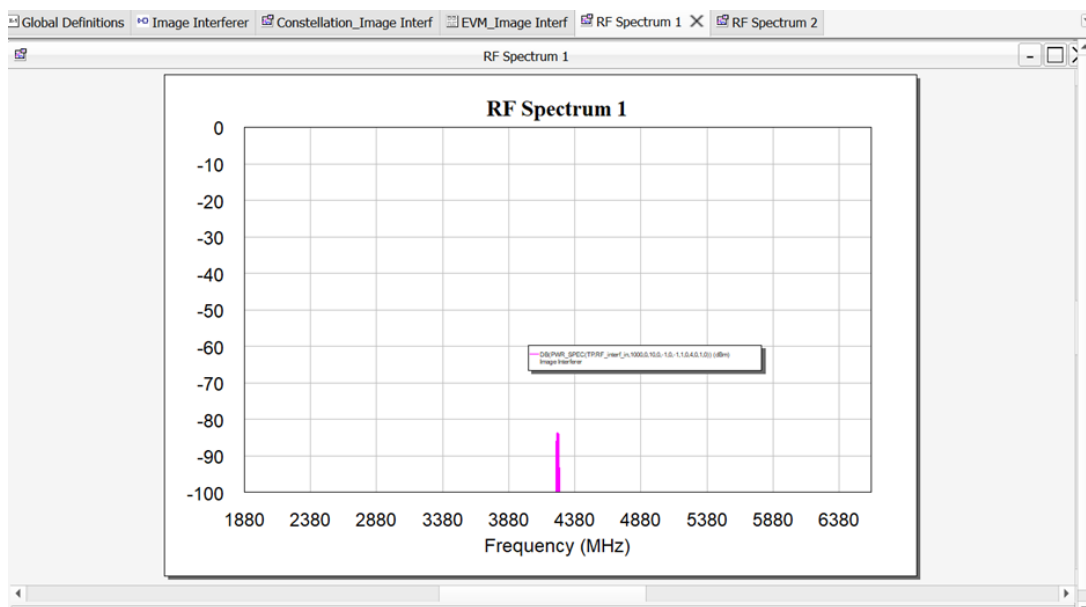


Figure: RF Spectrum 1

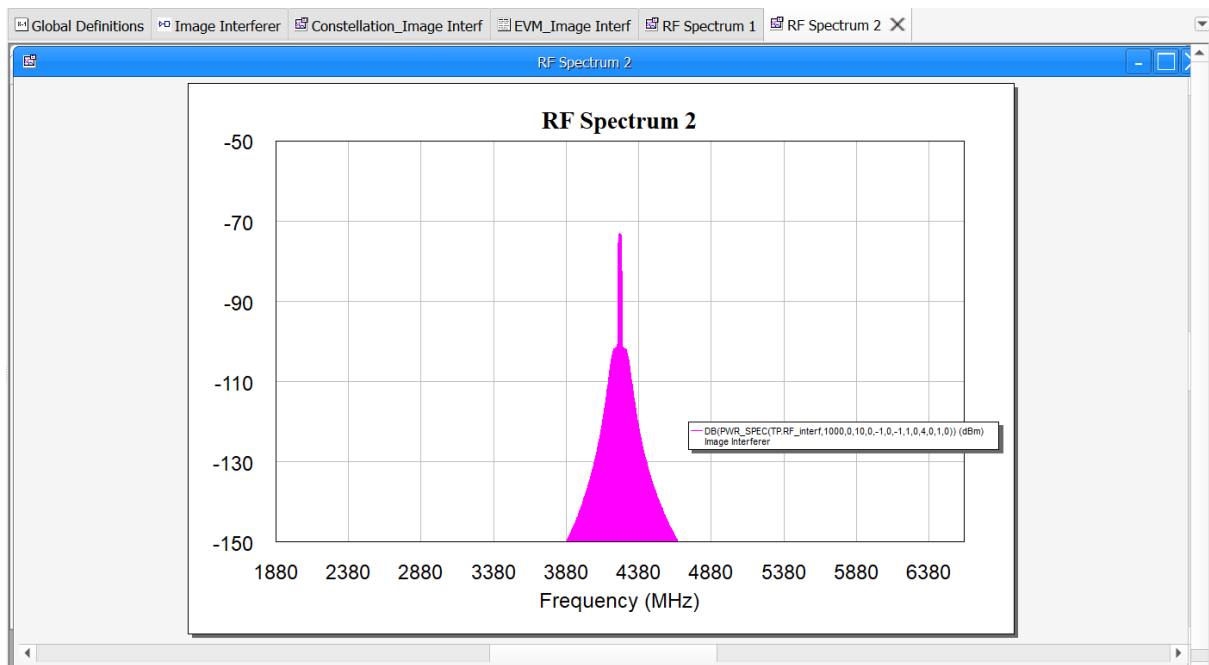


Figure : Spectrum 2